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RESEARCH ARTICLE

Controlling Doping Preference of Spiro-OMeTAD for Efficient and Stable Perovskite Solar Cells

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ABSTRACT

Organic–inorganic hybrid perovskite solar cells (PSCs) offer excellent optoelectronic properties and low-cost fabrication. However, their commercialization is limited by the instability and slow oxidation of spiro-OMeTAD induced by the hygroscopic Li-TFSI and highly volatile tBP dopants. In this work, two commercially available pyridine derivatives, 4-(dimethylamino)pyridine (DMAP) and 4-(trifluoromethyl)pyridine (TFMP), are introduced to regulate the Li^+–$TFSI^-$ environment and improve the doping behavior of spiro-OMeTAD. Spectroscopic analyses reveal that DMAP strongly interacts with Li^+ but exhibits negligible interaction with $TFSI^-$, thereby suppressing Li^+–$TFSI^-$ dissociation and hindering oxidation. In contrast, the electron-withdrawing TFMP promotes ion-pair dissociation through electrostatic interactions with $TFSI^-$, increasing the availability of free $TFSI^-$ required for efficient doping. As a result, TFMP-treated spiro-OMeTAD shows enhanced conductivity and an optimized work function, and TFMP also contributes to perovskite passivation through interactions with Pb^{2+}. Consequently, PSCs incorporating TFMP achieve a champion power conversion efficiency of 25.92% and exhibit markedly improved operational durability. Notably, under continuous maximum power point tracking (MPPT) at 1-sun illumination, TFMP-treated devices retained over 91% of their initial efficiency after 1000 h. These results demonstrate that TFMP not only effectively regulates dopants and accelerates the doping process but also enables excellent long-term and thermal stability compared to control devices.

1 | Introduction

Organic-inorganic hybrid perovskite solar cells (PSCs) have gained attention as next-generation solar cells capable of replacing silicon-based solar cells due to their low fabrication cost enabled by low-temperature and solution-based processes, tunable bandgap, and excellent electrical properties [1–3]. Starting with an efficiency report of 3.8% in 2009, PSCs have achieved remarkable progress, reaching 27.0% efficiency, nearing commercialization [4].

However, the commercialization of PSCs faces challenges due to their low device stability. Among the many components in PSCs, the hole-transporting layer (HTL) plays a crucial role in extracting and transporting the generated holes while effectively blocking electrons.

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One of the main issues that affects device stability is the instability caused by dopants introduced in commonly used hole-transporting materials (HTM) like 2,2',7,7'-Tetrakis(N,N-di-p-methoxyphenylamino)-9,9'-spirobifluorene (spiro-OMeTAD) [5, 6]. To improve the inherently low electrical conductivity and mobility of spiro-OMeTAD, p-type dopants such as bis(trifluoromethane)sulfonimide lithium salt (Li-TFSI) and 4-tert-butylpyridine (tBP) are typically employed. Li-TFSI enhances the hole transport properties by promoting the oxidation of spiro-OMeTAD, while tBP improves the solubility of Li-TFSI in solution and enhances hole extraction characteristics, making these dopants essential for spiro-OMeTAD [7, 8]. However, Li-TFSI and tBP possess properties that negatively impact the stability of PSCs. The hygroscopicity of Li-TFSI accelerates the decomposition of perovskite, severely impairing the long-term stability of the device [9]. tBP significantly reduces the glass transition temperature (Tg) of spiro-OMeTAD and has high volatility, causing voids on the HTL surface under high-temperature conditions, which is a major factor reducing the thermal stability of PSCs [10, 11]. Furthermore, the doping strategy of spiro-OMeTAD using Li-TFSI and tBP is highly sensitive to external environmental conditions due to the slow diffusion of oxygen into spiro-OMeTAD solutions [12, 13]. The time required for spiro-OMeTAD to achieve optimal work function (WF) and conductivity depends on doping efficiency, which directly impacts the device performance by influencing the resistance of the semiconductor thin film and charge recombination at the perovskite/HTL interface [14]. Collectively, the limitations of the Li-TFSI/tBP system arise from its high sensitivity to moisture, oxygen, and temperature, which destabilize both the dopants and the perovskite absorber.

To address these challenges, many researchers have conducted studies on novel dopants, dopant-controlling additives, and dopant-free HTMs. Gao's group reported spiro-OMeTAD^{2+}(TFSI$^-$)$_2$ and 4-tert-butyl-1-methylpyridinium bis(trifluoromethylsulfonyl)imide (TBMP$^+$TFSI$^-$) as alternatives to Li-TFSI and tBP [15]. Spiro-OMeTAD^{2+}(TFSI$^-$)$_2$ and TBMP$^+$TFSI$^-$ demonstrated high doping efficiency and device stability by rapidly oxidizing spiro-OMeTAD and adjusting the work function (WF), respectively. Bi's group reported 1,6-diazidohexane (N_3), which interacts simultaneously with Li-TFSI and tBP, suppressing the aggregation of Li-TFSI and the volatility of tBP [16]. The azide groups within the N_3 molecule not only stabilized the HTL thin film by suppressing the dopants but also facilitated the oxidation of spiro-OMeTAD, thereby accelerating doping. Additionally, numerous studies have reported polymers used as dopant-free HTMs, and devices based on these materials demonstrated high performance and excellent stability [17–19]. However, these novel materials have not been commercialized and remain feasible only at the laboratory scale due to technical barriers associated with their synthesis. Thus, there is a pressing need for dopant-engineering strategies that (i) maintain compatibility with commercially available HTMs and (ii) do not require the development of new and synthetically demanding dopants. In this study, commercially available additives, 4-(dimethylamino)pyridine (DMAP) and 4-(trifluoromethyl)pyridine (TFMP), were employed to address the instability and time-consuming oxidation process of spiro-OMeTAD. Both additives contain polarized pyridine rings, allowing them to modulate the Li$^+$–TFSI$^-$ interaction through Lewis basicity and local electrostatic effects. While DMAP, due to its strong electron-donating properties, significantly hindered the doping process of spiro-OMeTAD, TFMP demonstrated the ability to enhance doping efficiency through appropriate interaction with Li-TFSI. Consequently, TFMP improved the conductivity and WF of the HTL while significantly accelerating the doping process. Additionally, TFMP passivated the perovskite layer by interacting with Pb^{2+} without affecting the crystallinity of the perovskite. The optimized WF suppressed charge recombination within the device, leading to a significant improvement in V_{OC}. As a result, TFMP-treated PSCs not only achieved a champion PCE of 25.92% but also demonstrated superior long-term and thermal stability compared to the control devices.

2 | Results and Discussion

Figure 1 illustrates the molecular structures of the commercially available pyridine additives DMAP and TFMP. These two molecules were selected because their para-substituents provide opposite inductive and resonance effects, allowing systematic control over the electron density of the pyridine ring. DMAP contains a strongly electron-donating dimethylamino group, which significantly increases the electron density at the pyridinic nitrogen and makes it a strong Lewis base. In contrast, TFMP possesses a highly electron-withdrawing trifluoromethyl group, which renders the pyridine ring electron-poor and shifts its interaction preference toward anions. Owing to these substituent-driven electronic properties, both molecules exhibit resonance-induced dipole moments with distinct δ^+ and δ^- regions, enabling interactions with Li$^+$–TFSI$^-$ ion pairs in fundamentally different ways, as shown in Figure 1. By exploiting this contrasting behavior—DMAP is expected to coordinate strongly with Li$^+$, whereas TFMP—owing to its electron-poor ring—preferentially perturbs the electron-rich and highly polarizable TFSI$^-$ anion cloud—we aimed to regulate the Li-TFSI doping environment and mitigate instability induced by Li-TFSI and tBP, while simultaneously enabling faster oxidation of spiro-OMeTAD through controlled Li$^+$–TFSI$^-$ dissociation.

FTIR (Fourier Transform Infrared Spectroscopy) analysis (Figure 2) enables the evaluation of how each additive perturbs the local Li-TFSI environment, thereby modulating the availability of free TFSI$^-$ ions, which can ultimately influence the doping efficiency of spiro-OMeTAD. DMAP acts as a strong Lewis base due to the electron-donating nature of its –NMe_2 substituent, which increases the electron density on the pyridine nitrogen [20]. In contrast, TFMP functions as a relatively weak Lewis base because the strongly electron-withdrawing –CF_3 group lowers the electron density of the pyridine ring [21].

Ion pairs can be classified into three regimes according to the degree of interaction during the ionization–dissociation process [22]. A contact ion pair (CIP, type I) corresponds to direct Li$^+$–TFSI$^-$ association, whereas a solvent-separated ion pair (SSIP, type II) describes a state in which the two ions are slightly separated by one or more intervening solvent molecules. At full dissociation, the two species exist as free ions without direct Li$^+$–TFSI$^-$ interactions.

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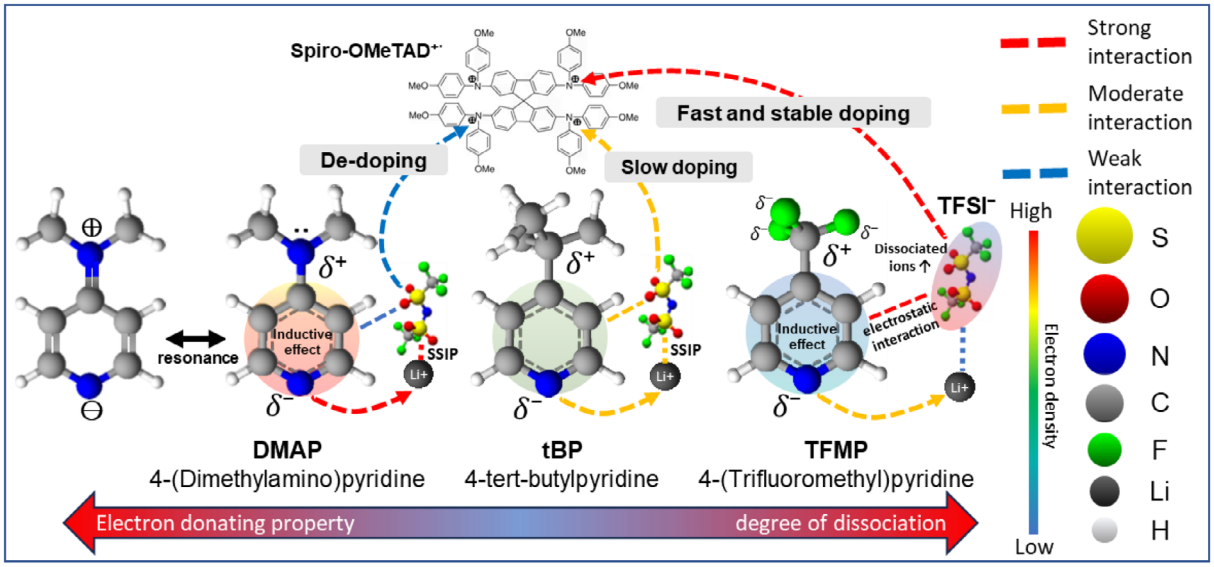

FIGURE 1 | Schematic illustration of how pyridine-based additives (DMAP, tBP, and TFMP) regulate the Li-TFSI doping environment and influence the oxidation behavior of spiro-OMeTAD. The degree of Li^+–$TFSI^-$ ion-pair dissociation is governed by the electron cloud distribution of each species and the resulting strength of their interactions. The strength of these interactions is represented by the color of the connecting lines (strong, moderate, and weak).

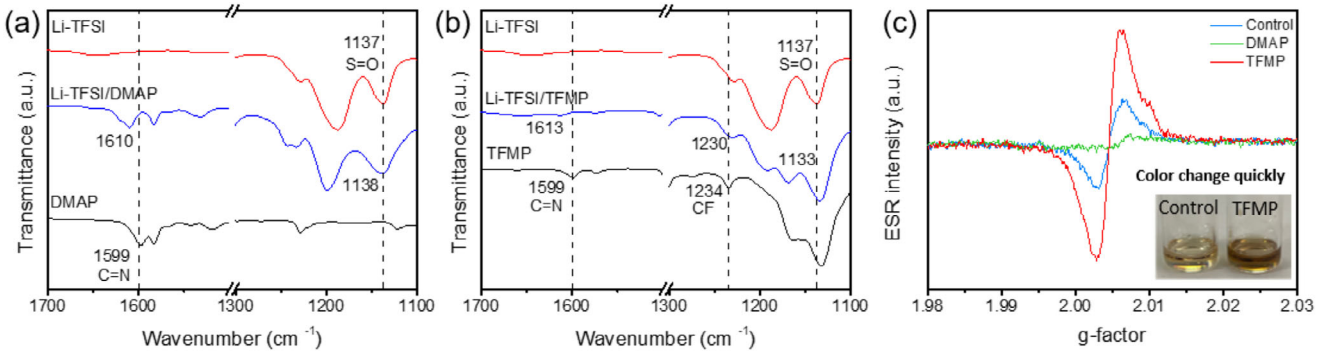

FIGURE 2 | FTIR spectra of (a) DMAP, Li-TFSI, and DMAP/Li-TFSI complex. (b) TFMP, Li-TFSI, and TFMP/Li-TFSI complex. (c) ESR signals of doped spiro-OMeTAD with and without the presence of DMAP and TFMP. The insert photo was taken 72 h after each solution was prepared.

The S═O stretching vibration of $TFSI^-$ appears at 1137 cm^{-1} in pristine Li-TFSI (Figure 2a). The negligible S═O shift (1137→1138 cm^{-1}) indicates that the $TFSI^-$ anion cloud remains largely unperturbed, consistent with a solvent-separated ion pair (SSIP) where Li^+–$TFSI^-$ coulombic association is only weakly altered [22].

In contrast, the incorporation of TFMP leads to a pronounced shift of the S═O band to 1133 cm^{-1} accompanied by noticeable broadening (Figure 2b). The strongly electron-withdrawing –CF_3 substituent of TFMP modifies the local electrostatic field of the pyridine ring, which can increase the electrostatic potential (ESP) and consequently alter its interaction with the highly polarizable $TFSI^-$ anion [23]. Such changes in the local electrostatic environment can induce a redistribution of the $TFSI^-$ electron cloud, resulting in the observed S═O vibrational shift. This perturbation also weakens the Li^+–$TFSI^-$ contact interaction, thereby increasing the fraction of free $TFSI^-$ species.

The C—F stretching vibration of TFMP also shifts from 1234 cm^{-1} in the pristine state to 1230 cm^{-1} upon mixing with Li-TFSI, accompanied by a noticeable change in intensity (Figure 2b). Although the fluorine atoms of the —CF_3 group possess very high electronegativity and therefore exhibit low affinity toward Li^+, this shift is likely induced indirectly by changes in the overall local environment created by TFMP in the Li-TFSI matrix.

It is noteworthy that the band at 1133 cm^{-1} lies within the typical vibrational range reported for CF_3 stretching modes in fluorinated aromatic systems [24]. Accordingly, the band observed near 1133 cm^{-1} in the TFMP-containing system can also be reasonably assigned to the symmetric C—F stretching mode of the —CF_3 group. Therefore, partial spectral overlap between TFMP C—F vibrations and the S═O stretching vibration of $TFSI^-$ cannot be excluded. Nevertheless, when compared with pristine Li-TFSI (1137 cm^{-1}), the TFMP/Li-TFSI mixture exhibits a clear red shift and band broadening in this region. Because the S═O vibrational frequency of $TFSI^-$ is sensitive to Li^+–$TFSI^-$ ion

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pairing, such red shifting behavior is consistent with perturbation of the Li–TFSI contact interaction. Within the CIP–SSIP–free ion framework described above, this behavior suggests that TFMP promotes partial ion-pair dissociation, increasing the fraction of more dissociated $TFSI^-$ species rather than forming strong direct coordination bonds.

The C═N stretching vibration of the pyridine ring appears at 1599 cm^{-1} for both pristine DMAP and TFMP (Figure 2a,b), reflecting a region of high electron density capable of interacting with the Li^+ component of the Li^+–$TFSI^-$ pair. Owing to its higher electron density, DMAP forms a stronger coordination bond with Li^+, whereas TFMP, with its electron-deficient ring, engages in comparatively weaker coordination. However, TFMP facilitates the dissociation of Li^+–$TFSI^-$ contact ion pairs, increasing the population of free Li^+ species. Importantly, this does not imply stronger intrinsic coordination between TFMP and Li^+. Rather, the increased degree of ion dissociation elevates the statistical frequency of transient Li^+–pyridine contacts arising from ion-pair perturbation.

Accordingly, upon mixing with Li-TFSI, the C═N stretching band shifts to 1610 cm^{-1} for DMAP and to 1613 cm^{-1} for TFMP.

Taken together, DMAP strongly coordinates with Li^+ but interacts only weakly, if at all, with $TFSI^-$, such that the Li^+–$TFSI^-$ interaction remains largely coulombic and predominantly in an SSIP-like state. In contrast, TFMP forms weaker coordination with Li^+ due to its lower Lewis basicity and reduced electron density, yet its electron-poor ring can perturb the electron-rich $TFSI^-$ anion through noncovalent electrostatic interactions. This perturbation promotes redistribution of the $TFSI^-$ electron cloud and facilitates the dissociation of Li^+–$TFSI^-$ contact ion pairs.

Overall, the observed changes in the S═O, C═N, and C—F vibrational modes collectively indicate that TFMP more effectively disrupts the Li-TFSI complex, leading to a higher fraction of free $TFSI^-$ species. Because free $TFSI^-$ plays a crucial role in stabilizing oxidized spiro-OMeTAD$^+$ during the doping process, the FT-IR results support that TFMP-containing compositions provide a more favorable ionic environment for rapid and efficient spiro-OMeTAD doping.

To further examine whether the additives interact with tBP, FTIR measurements were performed on tBP–additive mixtures (Figure S1). The C═N stretching vibration of pristine tBP appears at 1596 cm^{-1}. When mixed with DMAP (Figure S1a) and TFMP (Figure S1b), this band shifts slightly to 1597 cm^{-1} and 1599 cm^{-1}, respectively. These small yet detectable shifts indicate that both additives induce subtle perturbations in the local environment of tBP, suggesting the presence of weak interactions between tBP and each additive. Although these interactions are relatively weak, such subtle perturbations can still influence the local solvation and film-formation environment of spiro-OMeTAD-based hole-transporting layers. To further investigate the functional relevance of the TFMP–tBP interaction, the film formation behavior of spiro-OMeTAD layers was examined in the presence and absence of tBP (Figure S2). Notably, TFMP fails to form a continuous spiro-OMeTAD film in the absence of tBP, whereas a uniform and continuous HTL morphology is recovered when tBP is present. This observation suggests that tBP plays an essential role in stabilizing the film formation environment in TFMP-containing systems, implying a functional interplay between TFMP and tBP during HTL deposition. Previous studies have reported that tBP can alleviate solubility mismatch and suppress phase separation between spiro-OMeTAD and Li–TFSI, thereby enabling uniform HTL formation [8]. In this context, it is reasonable to infer that tBP may exert a similar stabilizing effect in TFMP-incorporated systems.

To examine the impact of the interaction between the pyridine derivatives and Li-TFSI on the oxidation process of spiro-OMeTAD, ESR (Electron Spin Resonance) analysis was conducted (Figure 2c). For this analysis, 100 mM of DMAP and TFMP were added to a mixture of spiro-OMeTAD, Li-TFSI, and tBP following the HTM solution recipe. Due to the significant influence of external conditions on the oxidation process, the analysis was performed immediately after the solution was prepared. The g-factor signals of all samples were consistent with the reported value of 2.0030 for the spiro-OMeTAD$^{+\cdot}$ radical generated during the oxidation process [25]. The ESR intensity of the DMAP-containing solution was significantly lower than that of the control, indicating that the oxidation of spiro-OMeTAD was substantially suppressed. This behavior aligns with the FTIR results showing that DMAP inhibits the dissociation of Li^+–$TFSI^-$ ion pairs, thereby reducing the concentration of free $TFSI^-$ available to stabilize oxidized spiro-OMeTAD. Moreover, the high electron density of the pyridine nitrogen in DMAP may additionally contribute to partial reduction of spiro-OMeTAD$^{+\cdot}$, leading to a more pronounced de-doping effect than that observed with tBP [26].

In contrast, the TFMP-containing solution exhibited more than twice the ESR intensity of the control. This enhancement is consistent with FTIR observations demonstrating that TFMP perturbs the Li-TFSI complex and facilitates the formation of free $TFSI^-$. The increased availability of free $TFSI^-$ promotes the stabilization of spiro-OMeTAD$^{+\cdot}$, resulting in improved doping efficiency. To further clarify the distinct roles of DMAP and TFMP in device operation, perovskite solar cells incorporating DMAP-doped spiro-OMeTAD were fabricated under identical conditions (Figures S3 and S4). The DMAP-based devices exhibited a reduced PCE of 17.50%, compared to 21.43% for the control and 22.23% for the TFMP-treated devices. This inferior performance is attributed to insufficient doping efficiency, leading to increased series resistance (R_s) and pronounced FF loss. These results are consistent with the ESR and FTIR analyses, which indicate that DMAP does not effectively promote radical formation nor sufficiently modulate the Li–TFSI dissociation environment. Therefore, subsequent experiments were conducted using TFMP as the sole additive.

As shown in Figure 3a, TFMP markedly accelerates the oxidation of spiro-OMeTAD. While both the control and TFMP-containing solutions were initially colorless, the control gradually developed a deep brown coloration over time due to the formation of oxidized spiro-OMeTAD [27]. In particular, the TFMP-containing solution exhibited a much faster color transition, suggesting rapid generation of radical cations. This accelerated oxidation was corroborated by the more prominent absorption peak at 520 nm in the UV–vis spectra, which is characteristic of the oxidized

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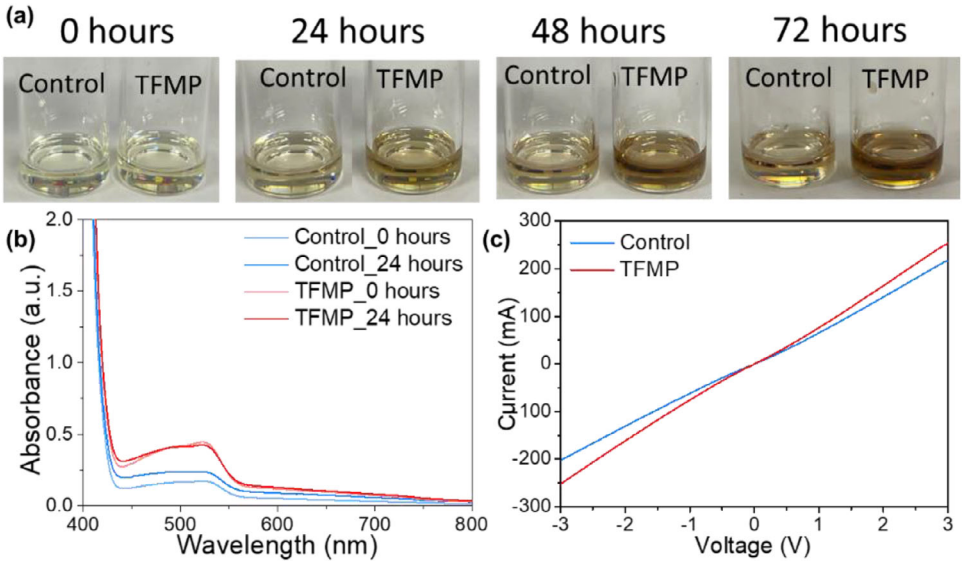

FIGURE 3 | (a) Images of the control and the TFMP-treated HTM solutions. (b) UV–vis absorption spectra comparing the control and the TFMP-treated HTM solution. (c) Conductivity measurements of spiro-OMeTAD films with and without TFMP treatment.

spiro-OMeTAD species. (Figure 3b) [28]. Specifically, while the peak intensity of the control solution gradually increased over time as oxidation proceeded, the TFMP-treated solution exhibited little change in intensity, suggesting that doping rapidly reached saturation within a short time after preparation. This observation is consistent with the time-dependent UV–vis analysis shown in Figure S5. The TFMP-treated solution completed rapid oxidation within the first few hours and remained stable thereafter, whereas the control solution exhibited a gradual increase over an extended period. Such rapid saturation behavior indicates that TFMP promptly establishes a stable doped state, effectively suppressing prolonged oxidation processes that can lead to over-doping or time-dependent instability in conventional spiro-OMeTAD systems.

To elucidate the dynamic oxidation process resulting from TFMP-induced modulation of the $TFSI^-$ coordination strength, and to indirectly compare changes in the associated energy barriers, time-dependent UV–vis measurements were conducted by monitoring the absorbance at 525 nm. The increase in absorbance was fitted using a single-exponential model according to the following equation:

$$\ln\left(\frac{A_{max} - A_0}{A_{max} - A_t}\right) = \mathrm{kt}$$

where A_{max} was defined as the absorbance value at 24 h to ensure consistent comparison within the same time window. Although oxidation in the control sample continues beyond this time range, the extracted k value represents an apparent rate constant for relative comparison under identical experimental conditions. To focus on the early-stage doping regime, where pseudo-first-order kinetic behavior is observed, kinetic fitting was performed using data within the initial 400 min.

The rate constants extracted from the linearized plots (Figure S6) were 0.01302 min^{-1} for TFMP and 0.00033 min^{-1} for the control sample. The substantially higher rate constant observed for TFMP confirms significantly accelerated dopant activation. Considering the Arrhenius relationship ($k \propto e^{-E_a/RT}$), a larger k value generally implies a lower effective activation barrier under identical experimental conditions. Therefore, the pronounced difference in the apparent k values suggests that TFMP lowers the energy barrier, thereby facilitating ion-pair dissociation and charge transfer.

To evaluate the impact of improved doping efficiency from TFMP treatment on the electrical properties of the HTL, conductivity measurements were performed by applying a voltage range of −3 V to 3 V to a device configured with an ITO/HTL/Au structure (Figure 3c). The HTL treated with TFMP demonstrated superior electrical properties, achieving a conductivity of 2.6×10^{-5} S cm^{-1}, compared to the control HTL, which showed a conductivity of 2.2×10^{-5} S cm^{-1}. In addition, as shown in Figure S7, time-dependent conductivity measurements revealed that the TFMP-containing HTL reached a stable conductivity value as early as the 12-h measurement, whereas the control HTL displayed a slower oxidation and doping process, with conductivity continuing to increase even until 24 h.

These results indicate that TFMP promotes rapid and stable doping of spiro-OMeTAD, enabling early establishment of a well-defined electronic state that is beneficial for both operational reliability and long-term stability. Consistent with the enhanced ESR intensity and accelerated oxidation kinetics discussed above, the faster stabilization of conductivity further supports the formation of a more favorable ionic environment for radical generation. The TFMP-induced shift in the Li–TFSI equilibrium toward a more dissociated state provides a quantitative link between ion dissociation and radical formation. The significantly higher apparent rate constant derived from time-dependent UV–vis analysis, together with the increased ESR signal intensity, indicates accelerated formation and stabilization of oxidized spiro-OMeTAD species. The faster conductivity stabilization further supports that TFMP creates a more favorable ionic environment for efficient dopant activation.

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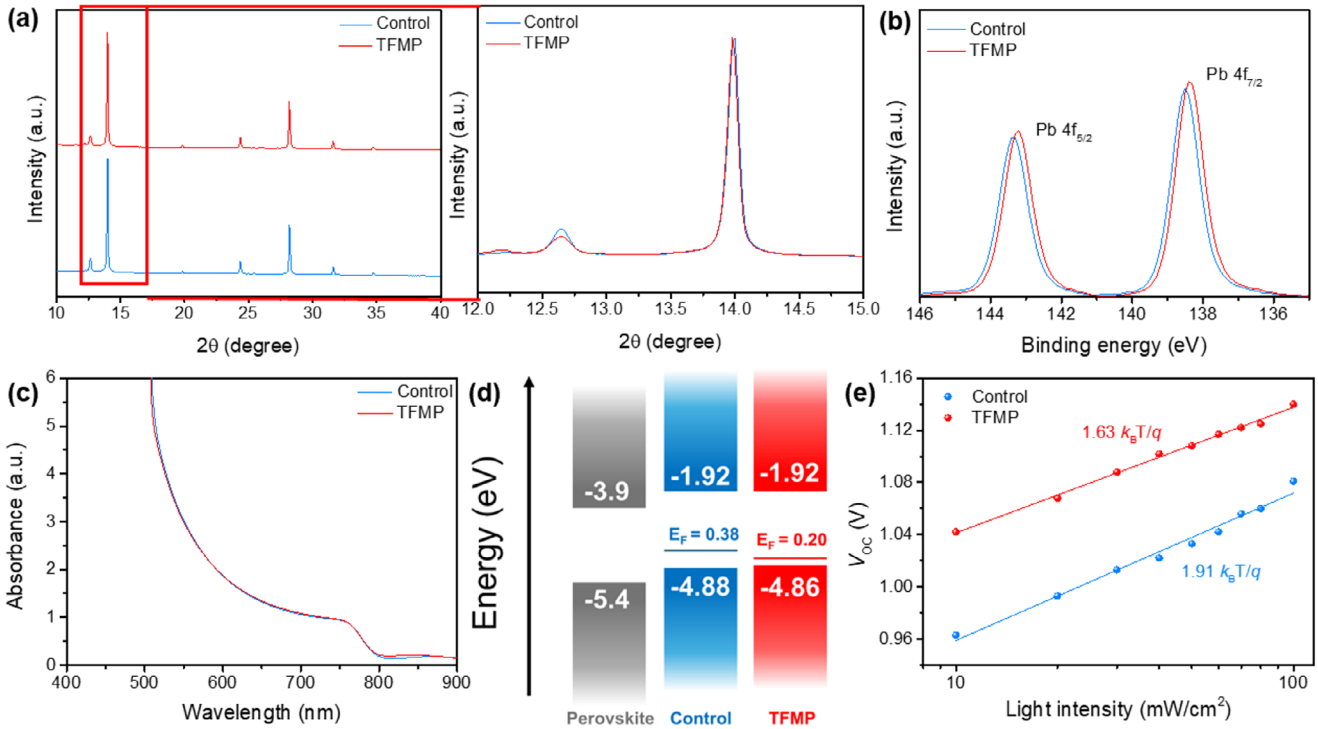

FIGURE 4 | (a) XRD spectra of the perovskite film with and without TFMP treatment. (b) XPS spectra showing the Pb 4f peaks for the control and TFMP-treated perovskite films. (c) UV–vis spectra of the perovskite film with and without TFMP treatment. (d) Energy level diagram comparing the perovskite, control, and TFMP-treated HTL. (e) V_{OC} as a function of light intensity for devices based on control and TFMP-treated PSCs.

To determine whether TFMP affects the perovskite structure, XRD measurements were performed after HTL deposition and CB washing (Figure 4a). Both the control and TFMP-treated samples exhibited characteristic perovskite peaks at $2\theta = 13.9°$ and 28.2°, confirming that TFMP does not alter the perovskite lattice structure and crystallinity. In contrast, the intensity of the peak at $2\theta = 12.6°$, associated with PbI_2 formation caused by perovskite degradation was reduced in the TFMP-treated sample. This suggests that F in TFMP interacts with surface Pb^{2+} sites and contributes to defect passivation. [29]

As shown in Figure 4b, XPS measurements were conducted to further examine the passivation effect. The Pb 4f spectrum of the TFMP-treated perovskite film exhibited a shift toward lower binding energy compared to the control sample, indicating increased electron density around the Pb atoms. Such a shift can be attributed to coordination interactions between surface Pb^{2+} species and electron-rich Lewis base functionalities in TFMP [30]. This electronic interaction is consistent with the reduced PbI_2 signal observed in the XRD results, supporting the possibility of surface defect passivation at Pb-related sites. UV–vis measurements of the perovskite/HTL absorption exhibited comparable spectra, indicating that TFMP did not influence the energy levels or lattice structure of the perovskite (Figure 4c).

Figure 4d presents a diagram of the HOMO and WF levels of the control and TFMP-treated HTL, derived from UPS measurements (Figure S8). The HOMO levels for the control and TFMP-treated samples were nearly identical, measured at −4.88 eV and −4.86 eV, respectively. However, the TFMP-treated spiro-OMeTAD exhibited a slightly deeper Fermi level (EF), positioning it closer to the HOMO level compared to the control. This indicates a higher degree of oxidation consistent with the enhanced doping observed in the FTIR and ESR results. Since the HOMO level of the HTL, which is higher than the valence band (VB) of the perovskite, has minimal influence on the V_{OC} of the device, this slight shift is unlikely to affect V_{OC} significantly [31]. Within an appropriate HOMO energy range, an HTL with an optimal WF that is well-aligned with the quasi-Fermi level (QFL) of the perovskite can effectively reduce voltage loss and enhance the V_{OC} of the device [32]. The enhancement in WF achieved through TFMP, reflecting its enhanced doping effect, lowers the hole extraction barrier between the perovskite and the TFMP-treated HTL, effectively reducing non-radiative recombination at the interface [15]. Consistent with this, V_{OC} of TFMP-treated PSCs was markedly higher than that of the control devices (Figure 4e). Additionally, the TFMP-treated devices exhibited a lower ideality factor, indicating reduced recombination and leakage current within the optoelectronic system [33].

It was confirmed that TFMP enhanced the doping efficiency and electrical properties of spiro-OMeTAD without impacting the perovskite lattice. Consequently, optimization of the additive concentration was carried out to incorporate TFMP into the device. As illustrated in Figure S9. When TFMP was incorporated into the device at concentrations below 200 mM, the highest performance was achieved at 100 mM. This concentration aligned with that used in the film characterization experiments. Consequently, 100 mM was identified as the optimal concentration, as it enhanced the electrical properties of the HTL and improved overall device performance when incorporated into spiro-OMeTAD.

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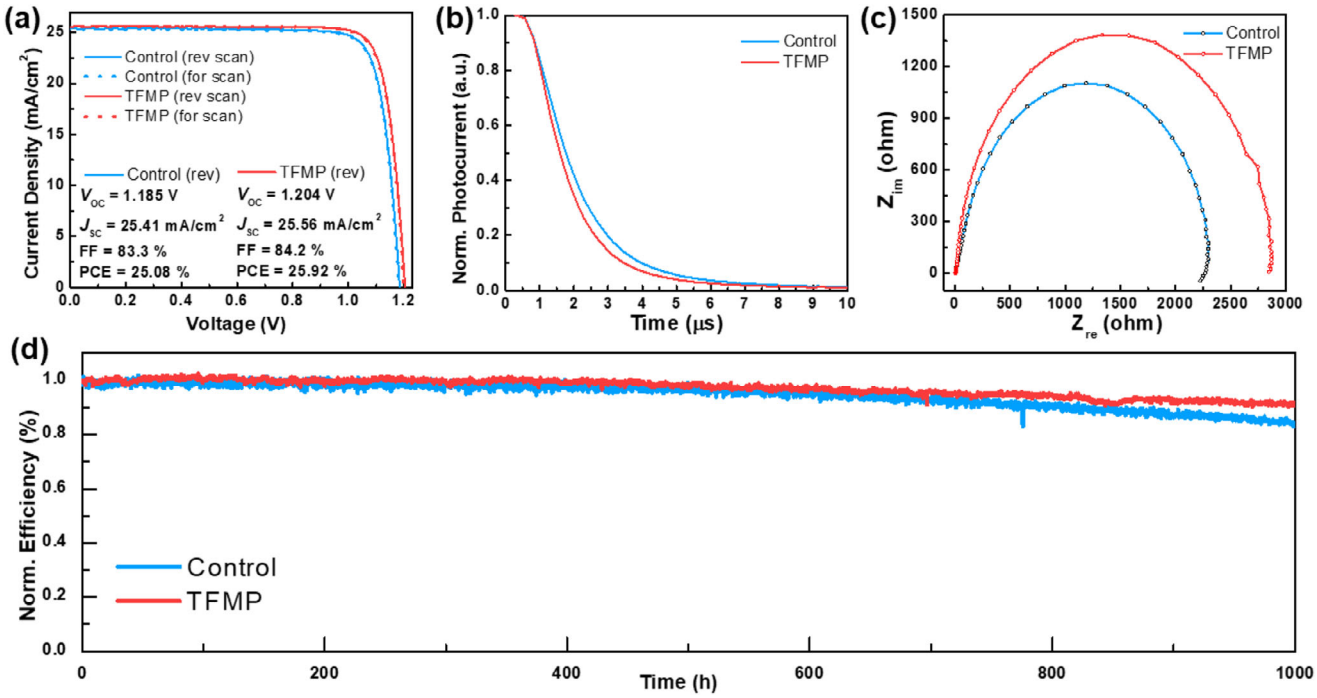

FIGURE 5 | (a) *J–V* curves of control and TFMP champion PSCs. (b) Transient photocurrent (TPC) decay curves of control and TFMP PSCs. (c) Nyquist plots obtained from electrochemical impedance spectroscopy (EIS) measurements of control and TFMP PSCs. (d) Normalized efficiencies of the encapsulated devices as a function of time under continuous 1-sun operation in air (relative humidity: 30–40%, temperature: 30–40°C) at their maximum power point (MPP). The initial efficiencies were 23.49% (control) and 24.23% (TFMP-treated).

To evaluate the impact of the enhanced doping efficiency of spiro-OMeTAD at the optimal TFMP concentration on the performance of optoelectronic devices, PSCs were fabricated. Figure S10a,b displays the *J–V* curves of the devices measured at different times after fabrication. The control device showed an increase in V_{OC} and FF over time, reflecting improvements in the WF and conductivity of the HTL as the doping process progressed. In contrast, the TFMP-treated PSCs showed almost unchanged efficiencies between 12 h and 209 h, demonstrating that TFMP rapidly brings spiro-OMeTAD to its fully oxidized state and, importantly, preserves this doping configuration by suppressing dopant redistribution or de-doping, thereby enabling long-term operational stability.

To further optimize the device performance, the TFMP-modified spiro-OMeTAD developed in this study was applied on top of the FTO/SnO_2/perovskite/OABr architecture. Figure 5a, Figure S11a, and Table S1 present the PCE distribution and champion performance of the TFMP device incorporating TFMP-modified spiro-OMeTAD, compared to the control device employing unmodified spiro-OMeTAD. Owing to the defect passivation capability of TFMP, the average V_{OC} increased from 1.18 V (control) to 1.20 V (TFMP), and the *FF* improved from 82.45% to 83.76%, attributed to enhanced conductivity. Consequently, the PCE of the champion device was enhanced from 25.08% (control) to 25.92% (TFMP), and a 100-s maximum power point tracking (MPPT) analysis further confirmed the improved performance, with efficiency increasing from 24.9% (control) to 25.9% (TFMP), as shown in Figure S11b. The external quantum efficiency (EQE) spectra of the control and TFMP devices exhibited integrated Jsc values of 24.76 mA cm^{-2} and 24.89 mA cm^{-2}, respectively, which closely matched the average *J*sc difference observed between the control and TFMP devices (Figure 5a; Figure S11c).

To exclude any effects other than those originating from the TFMP-modified spiro-OMeTAD developed in this study, device stability was evaluated using a device structure without additional perovskite surface modification employed for device optimization. Under continuous 1-sun operation at the MPP, the TFMP device retained over 91% of its initial efficiency after 1000 h, whereas the control device exhibited relatively poor stability, maintaining only about 83% of its initial efficiency. (Figure 5d; Figure S11d) Although the absolute difference in efficiency retention between the two devices is on the order of ~10 percentage points, this comparison is made in a high-efficiency regime where the initial power conversion efficiency already exceeds 25%, and further improvements in operational stability are generally incremental [34, 35].

To further support the enhancement in device performance, transient photocurrent (TPC) measurements under 0.3-sun illumination and electrochemical impedance spectroscopy (EIS) at a bias voltage of approximately 1 V in the dark were conducted (Figure 5b,c). Compared to the control, the shorter TPC lifetime observed in the TFMP device further confirms the enhanced charge carrier extraction to the electrode, attributed to improved conductivity [36]. Additionally, the recombination resistance (R_{rec}) of the TFMP device was 2.81 kΩ, higher than that of the control device (2.06 kΩ), while the series resistance (R_s) slightly decreased, further indicating suppressed charge recombination and enhanced carrier transport throughout the device [37].

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To evaluate whether the effect of TFMP extends to other triphenylamine-based HTMs, we additionally tested PTAA (Figures S12 and S13). The PTAA-based PSCs also exhibited an efficiency improvement, increasing from 24.42% to 25.11% upon TFMP treatment.

We conducted additional experiments to evaluate whether the TFMP additive is also applicable to poly(3-hexylthiophene) (P3HT), another widely used organic hole-transport material. Because P3HT can exhibit hole-transport characteristics even without dopants such as Li-TFSI, it is often used without Li-TFSI [38]; however, in some studies, Li-TFSI has also been incorporated to further enhance hole-transport performance [39]. Therefore, we prepared hole-transport layers using P3HT solutions with and without Li-TFSI and fabricated perovskite solar cells (PSCs) based on these layers. In both cases, incorporating TFMP resulted in notable improvements in V_{OC} and FF. (Figure S14). The performance enhancement observed in the P3HT-based devices without Li-TFSI appears to originate from the defect-passivation effect of TFMP on the perovskite photoactive layer, as discussed above. For P3HT-based devices with Li-TFSI, the additional performance improvement appears to arise from not only TFMP's defect-passivation effect on the perovskite photoactive layer but also its regulation of Li-TFSI–related doping chemistry. Overall, these results further support the effectiveness and broader applicability of the TFMP proposed in this study.

While dopant-free HTMs have recently attracted significant attention due to their chemical simplicity and improved intrinsic stability by eliminating hygroscopic dopants, recent studies emphasize that high device performance is typically achieved through careful molecular design, optimized energy-level alignment, and interfacial engineering strategies [40]. In many cases, additional control over perovskite crystallization or interface modification is employed to compensate for the relatively lower intrinsic conductivity of dopant-free systems. In this context, the present results highlight the practical relevance of stabilizing doped HTMs—such as spiro-OMeTAD, which remains widely used in high-efficiency perovskite solar cells—through a simple and broadly applicable additive strategy that does not require the development of new transport materials.

A comparative summary of representative high-performance additives and dopant-free HTMs reported in the literature is provided in Table S2. This comparison further highlights that the present strategy maintains compatibility with commercially available HTMs and the conventional Li–TFSI-based doping system, without requiring synthetically complex dopants or the development of entirely new charge-transport materials. Notably, the TFMP-based devices achieved a power conversion efficiency of 25.92%, placing them among the higher-performing additive- or dopant-engineered perovskite solar cells reported to date. These results underscore the practical effectiveness of this simple additive strategy and its potential for scalable implementation.

Additionally, to assess the long-term stability of the control and TFMP-treated devices, PCE values were monitored using the *J*–*V* curves of unencapsulated devices stored under ambient conditions of 25–35°C and 10% relative humidity (Figure 6a). The TFMP-treated PSCs retained over 90% of their initial efficiency for 1800 h, whereas the control devices showed an initial increase followed by a rapid decline. In particular, the pronounced performance increase observed during the initial ~200 h for the control device can be attributed to the gradual establishment of the optimal doping state. As evidenced by the time-dependent UV–vis and conductivity analyses discussed above, the control HTL undergoes progressive oxidation, leading to increased radical cation formation and enhanced hole transport over time. This suggests that the control device is initially in a relatively underdoped state, and its performance improves as oxidation proceeds toward an optimal doping level.

In contrast, the TFMP-treated HTL reaches a near-saturated doping state within a short time after preparation, leaving limited room for further performance enhancement during early-stage operation. Consequently, the TFMP device exhibits minimal initial efficiency evolution while maintaining a more stable doped state throughout prolonged operation. The subsequent rapid degradation of the control device is consistent with the slow oxidation and eventual over-oxidation of spiro-OMeTAD in the conventional Li-TFSI/tBP system.

As previously discussed, tBP acts as a plasticizer that significantly lowers the Tg of spiro-OMeTAD. To examine whether TFMP, whose molecular framework resembles that of tBP, affects the intrinsic thermal stability of spiro-OMeTAD, differential scanning calorimetry (DSC) measurements were performed (Figure S15). The spiro-OMeTAD powders were prepared using the standard HTM recipe and subsequently dried in a vacuum oven for 24 h. The glass transition temperatures (Tg) of the control and TFMP-treated spiro-OMeTAD were measured to be 122.5°C and 122.7°C, respectively, indicating that TFMP does not alter the inherent thermal properties of the HTM. Although Tg remains unchanged, TFMP is expected to improve the high-temperature stability of spiro-OMeTAD by maintaining a more stable Li-TFSI doping environment, thereby mitigating dopant redistribution or de-doping processes that typically occur under thermal stress. Thermal stability tests conducted at 65°C under an N_2 atmosphere revealed that TFMP-treated PSCs retained 80% of their initial efficiency for approximately 1200 h, whereas the control devices experienced a rapid efficiency drop, resulting in over twice the operational lifetime for the TFMP-treated device under identical conditions (Figure 6b). These results demonstrate that TFMP not only effectively regulates dopants, significantly reducing the doping process time, but also enhances both long-term and thermal stability.

3 | Conclusions

In this study, we investigated two commercially available pyridine-based additives, DMAP and TFMP, to regulate the Li–TFSI doping environment of spiro-OMeTAD. FTIR and ESR analyses reveal that these additives induce fundamentally distinct effects on the Li^+–$TFSI^-$ ion pair. DMAP, owing to its strong Lewis basicity, preferentially coordinates with Li^+ and suppresses Li^+–$TFSI^-$ dissociation, thereby hindering the oxidation of spiro-OMeTAD. In contrast, TFMP perturbs the electron-rich $TFSI^-$ anion cloud through electrostatic interactions and promotes the formation of free $TFSI^-$, enabling more efficient oxidation and significantly enhancing the conductivity of the HTL. UPS analysis confirms that TFMP-treated spiro-OMeTAD exhibits a

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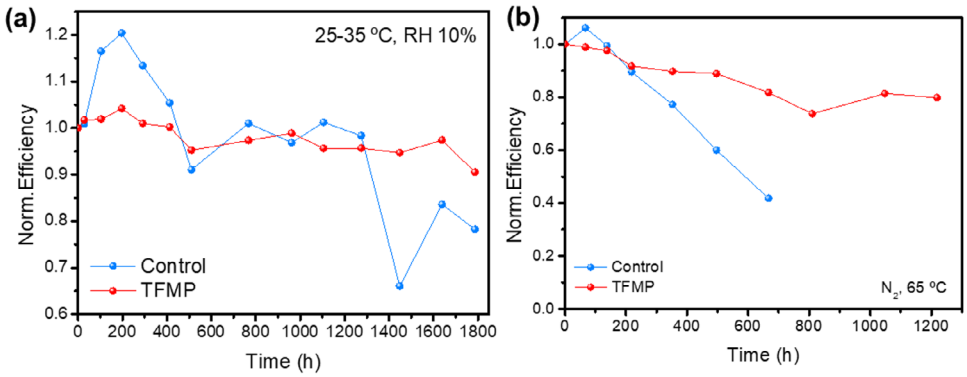

FIGURE 6 | (a) Operational stability of the devices under ambient conditions at 25°C with 10% relative humidity. The initial efficiencies were 17.51% (control) and 21.60% (TFMP-treated). (b) Thermal stability of the devices in a nitrogen atmosphere at 65°C. The initial efficiencies were 18.63% (control) and 22.23% (TFMP-treated).

more favorable work function, improving energetic alignment with the perovskite and reducing the hole-extraction barrier. Structural analyses further show that TFMP does not modify the perovskite lattice, while XRD and XPS results suggest that TFMP contributes to defect passivation through interactions with surface Pb^{2+} species, thereby mitigating interfacial recombination. DSC measurements indicate that TFMP does not alter the intrinsic thermal properties of spiro-OMeTAD. However, by stabilizing the Li–TFSI doping environment, TFMP is expected to mitigate thermally induced dopant redistribution or de-doping, contributing to improved high-temperature durability of the HTL.

Device-level characterization further confirms the performance advantages of TFMP-modified spiro-OMeTAD. Time-dependent *J–V* measurements reveal that TFMP enables rapid dopant activation, yielding stable efficiencies within 12 h, unlike the gradually improving control. TFMP incorporation increases the average V_{OC} (1.18 → 1.20 V) and FF (82.45 → 83.76%), leading to a higher champion PCE of 25.92%. EQE spectra verify the slight J_{SC} improvement, consistent with the integrated photocurrent values. Operational long-term stability is markedly enhanced, with TFMP-based PSCs retaining over 91% of their initial efficiency after 1000 h of MPP tracking, compared with ~83% for the control. In addition, TPC and EIS analyses reveal faster charge extraction, higher recombination resistance (2.81 vs 2.06 kΩ), and lower series resistance, collectively demonstrating reduced interfacial recombination and improved carrier transport throughout the device.

Overall, this work demonstrates that additive-mediated modulation of ion–dopant interactions is an effective strategy for improving the performance and stability of spiro-OMeTAD-based PSCs, and establishes TFMP as a practical and efficient additive for accelerating doping and enhancing device reliability.

4 | Experimental Section/Methods

4.1 | Materials

Ethanol, acetone, acetonitrile, isopropanol, chlorobenzene, dimethyl sulfoxide (DMSO), dimethylformamide (DMF), $SnCl_2$ dihydrate, thioglycolic acid (TGA), urea, 4-*tert*-butylpyridine (*t*BP), and bis(trifluoromethane)sulfonimide lithium salt (Li-TFSI) were purchased from Sigma-Aldrich. Tin(IV) oxide colloidal dispersion was purchased from Alfa Aesar. Formamidinium iodide (FAI), methylammonium bromide (MABr), octylammonium bromide (OABr), and cobalt(III) tris(bis(trifluoromethylsulfonyl)imide) (FK209) were purchased from Greatcell Solar Materials. Lead(II) iodide (PbI_2) and lead(II) bromide ($PbBr_2$) were purchased from TCI. 2,2´,7,7´-tetrakis(*N,N´*-di-p-methoxyphenylamine)-9,9´-spirobifluorene (spiro-OMeTAD) was purchased from Lumtec. HCl was purchased from Samchun. Polytriarylamine (PTAA) was purchased from MS Solutions. Poly(3-hexylthiophene) (P3HT) was purchased from 1-Material.

4.2 | Solar Cell Fabrication

ITO glass substrates were sequentially cleaned using deionized water and isopropanol through ultrasonic treatment, with each step lasting 10 min. Following cleaning, the substrates were dried and exposed to UV-ozone treatment for 15 min. A SnO_2 solution, prepared by diluting the commercial SnO_2 dispersion in water at a 1:5 v/v ratio, was spin-coated onto the cleaned ITO substrates at 3000 rpm for 30 s and then thermally annealed at 100°C for 30 min. After cooling to room temperature, the substrates underwent an additional 15 min of UV-ozone treatment. A perovskite precursor solution was formulated by combining FAI (1.27 M), PbI_2 (1.4 M), MABr (0.06 M), $PbBr_2$ (0.06 M), and MACl (0.45 M) in a solvent mixture of DMF and DMSO (8:1 v/v). This solution was spin-coated onto the SnO_2-coated substrates using a two-step program: 1000 rpm for 5 s and 5000 rpm for 20 s. During the final step, 1 mL of diethyl ether was added dropwise to the rotating substrate within the last 10 s of the spin-coating process. For the fabrication of the optimized device, SnO_2 was deposited onto FTO glass (Asahi) via the chemical bath deposition (CBD) method. The perovskite layer was prepared using $FAPbI_3$ (1.5 M) with 4% $MAPbBr_3$ additive and coated following the same procedure described above. Subsequently, an OABr solution (10 mM in IPA) was spin-coated onto the formed perovskite layer at 5000 rpm for 30 s and then annealed at 100°C for 13 min. Subsequently, a spiro-OMeTAD solution (100.0 mg mL^{-1} in chlorobenzene) containing 23 μL of Li-TFSI (540 mg mL^{-1} in acetonitrile), 10 μL of FK209 (370 mg mL^{-1} in

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acetonitrile), 39 μL of tBP, and 100 mM TFMP was applied onto the perovskite layer by spin-coating at 3000 rpm for 30 s. To apply TFMP to PTAA hole-transport materials, we prepared a PTAA solution by adding 100 mM TFMP to the PTAA hole-transport solution prepared according to our previous study [41]. Then, the solution was spin-coated on the perovskite layer at 3000 rpm for 30 s to form a PTAA hole-transport layer. To incorporate TFMP into P3HT, two P3HT solutions—one containing Li-TFSI and tBP and the other without these additives—were prepared according to the literature [39, 42]. TFMP was then added to each solution at a concentration of 100 mM to obtain the TFMP-containing P3HT solutions. The prepared P3HT solutions were spin-coated onto the perovskite layer at 3000 rpm for 30 s to fabricate P3HT-based hole-transport layers. Finally, metallic electrodes (100 nm Ag and 80 nm Au) were deposited onto the substrate using thermal evaporation under high vacuum. The device stability evaluation was conducted without applying additional perovskite surface treatments (e.g., OABr passivation), which are commonly used to maximize the initial power conversion efficiency, in order to isolate the effect of TFMP. For the light stability test shown in Figure 5d, encapsulated devices were used, whereas unencapsulated devices were employed for Figure S5b, Figure 6a, and Figure 6b. Although OABr passivation can enhance device efficiency, it has been reported to influence device stability; therefore, it was intentionally omitted in this study [43]. Accordingly, stability measurements were performed on devices with the structure FTO/SnO_2/perovskite/spiro-OMeTAD (with or without TFMP)/Au.

4.3 | Characterization

Device Characterization: Solar cell characterization was performed in air using a Keithley 2450 source measurement unit (SMU) and an Oriel xenon lamp (450 W) equipped with an AM1.5 filter. The illumination conditions were set to AM 1.5G at 100 mW/cm^2, calibrated with a silicon reference cell certified by Newport (MKS Instruments/Newport PV lab). Current density–voltage (*J–V*) curves for all devices were recorded at a scan rate of 0.1 V/s. The active device area was defined using a dark mask with an aperture of 0.0937 cm^2 (0.0804 cm^2 for the champion device). EQE measurement was conducted using a quantum efficiency measurement system (QuantX-300, Newport).

UV–vis Measurement: The absorbance of the film and solution was measured using ultraviolet-visible spectroscopy (UV–vis) with a V-770 spectrophotometer from Jasco Inc.

XPS and UPS Measurement: X-ray photoelectron spectroscopy (XPS) and ultraviolet photoelectron spectroscopy (UPS) were performed using a NEXSA G2 system from Thermo to investigate surface chemical states and energy levels.

XRD Measurement: To verify the crystal structure, X-ray diffraction (XRD) was measured using a SmartLab 9 kW diffractometer of Rigaku Corporation.

Conductivity Measurement: The conductivity of the HTL was evaluated by applying a voltage range of −2 V to 2 V on a device configured with an ITO/HTL/Au structure.

ESR Measurement: Electron spin resonance (ESR) measurements were performed using an EMXplus-9.5/12/P/L system (Bruker, Germany) with the HTM solution.

DSC Analysis: Differential Scanning Calorimeter (DSC, N-650, Scinco, Korea) was utilized to determine the glass transition temperature (Tg) of spiro-OMeTAD under a nitrogen atmosphere, with a heating rate of 10°C/min over a temperature range of 25–300°C.

TPC and EIS Measurements: Transient photocurrent (TPC) measurements were performed using a characterization platform (Paios, Fluxim). Nyquist plots were obtained using a Potentio/Galvanostat-Impedance spectroscopy system (Autolab, Metrohm).

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Author Contributions

Yeji Kim and Kyung Mun Yeom contributed equally to this work and carried out all the experiments, including material preparation, characterization, and electrochemical analysis. Seulki Song supervised this project and was responsible for the writing, reviewing, and editing. All authors discussed the experiments and results. All authors have given approval for the final version of the manuscript.

Conflicts of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

1. H. Kim, B. Noh, C. B. Lee, et al., "Nanoscopic Parylene Layer: Enhancing Perovskite Solar Cells Through Parylene-D Passivation," *Small Methods* 10 (2025): 2500395, https://doi.org/10.1002/smtd.202500395.

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2. N.-G. Park, "Perovskite Solar Cells: An Emerging Photovoltaic Technology," *Materials Today* 18, no. 2 (2015): 65–72, https://doi.org/10.1016/j.mattod.2014.07.007.

3. J. Y. Kim, J.-W. Lee, H. S. Jung, H. Shin, and N.-G. Park, "High-Efficiency Perovskite Solar Cells," *Chemical Reviews* 120, no. 15 (2020): 7867–7918, https://doi.org/10.1021/acs.chemrev.0c00107.

4. (n.d.) "Best Research-cell Efficiency Chart," National Renewable Energy Laboratory (NREL), https://www.nrel.gov/pv/cell-efficiency.html, (accessed: November 2025).

5. J. Park, J. Kim, H.-S. Yun, et al., "Controlled Growth of Perovskite Layers with Volatile Alkylammonium Chlorides," *Nature* 616, no. 7958 (2023): 724–730, https://doi.org/10.1038/s41586-023-05825-y.

6. J. J. Yoo, G. Seo, M. R. Chua, et al., "Efficient Perovskite Solar Cells via Improved Carrier Management," *Nature* 590, no. 7847 (2021): 587–593, https://doi.org/10.1038/s41586-021-03285-w.

7. A. Abate, T. Leijtens, S. Pathak, et al., "Lithium Salts as "Redox Active" p-Type Dopants for Organic Semiconductors and Their Impact in Solid-state Dye-Sensitized Solar Cells," *Physical Chemistry Chemical Physics* 15, no. 7 (2013): 2572, https://doi.org/10.1039/C2CP44397J.

8. S. Wang, M. Sina, P. Parikh, et al., "Role of 4-tert-butylpyridine as a Hole Transport Layer Morphological Controller in Perovskite Solar Cells," *Nano letters* 16, no. 9 (2016): 5594–5600, https://doi.org/10.1021/acs.nanolett.6b02158.

9. G.-W. Kim, G. Kang, J. Kim, et al., "Dopant-free Polymeric Hole Transport Materials for Highly Efficient and Stable Perovskite Solar Cells," *Energy & Environmental Science* 9, no. 7 (2016): 2326–2333, https://doi.org/10.1039/C6EE00709K.

10. J. Zhang, T. Zhang, L. Jiang, U. Bach, and Y.-B. Cheng, "4-tert-Butylpyridine Free Hole Transport Materials for Efficient Perovskite Solar Cells: A New Strategy to Enhance the Environmental and Thermal Stability," *ACS Energy Letters* 3, no. 7 (2018): 1677–1682, https://doi.org/10.1021/acsenergylett.8b00786.

11. T. Malinauskas, D. Tomkute-Luksiene, R. Sens, et al., "Enhancing Thermal Stability and Lifetime of Solid-State Dye-sensitized Solar Cells via Molecular Engineering of the Hole-transporting Material Spiro-OMeTAD," *ACS applied materials & interfaces* 7 (2015): 11107–11116, https://doi.org/10.1021/am5090385.

12. L. Ye, J. Wu, S. Catalán-Gómez, et al., "Superoxide Radical Derived Metal-Free Spiro-OMeTAD for Highly Stable Perovskite Solar Cells," *Nature Communications* 15, no. 1 (2024): 7889, https://doi.org/10.1038/s41467-024-52199-4.

13. J. Kong, Y. Shin, J. A. Röhr, et al., "CO_2 doping of Organic Interlayers for Perovskite Solar Cells," *Nature* 594, no. 7861 (2021): 51–56, https://doi.org/10.1038/s41586-021-03518-y.

14. M. Stolterfoht, P. Caprioglio, C. M. Wolff, et al., "The Impact of Energy Alignment and Interfacial Recombination on the Internal and External Open-circuit Voltage of Perovskite Solar Cells," *Energy & environmental science* 12, no. 9 (2019): 2778–2788, https://doi.org/10.1039/C9EE02020A.

15. T. Zhang, F. Wang, H.-B. Kim, et al., "Ion-Modulated Radical Doping of Spiro-OMeTAD for More Efficient and Stable Perovskite Solar Cells," *Science* 377, no. 6605 (2022): 495–501, https://doi.org/10.1126/science.abo2757.

16. Y. Han, G. Zhang, H. Xie, et al., "Azide Additive Acting as a Powerful Locker for Li+ and TBP in Spiro-OMeTAD Toward Highly Efficient and Stable Perovskite Solar Cells," *Nano Energy* 96 (2022): 107072, https://doi.org/10.1016/j.nanoen.2022.107072.

17. Y. Bai, Z. Zhou, Q. Xue, et al., "Dopant-Free Bithiophene-Imide-Based Polymeric Hole-Transporting Materials for Efficient and Stable Perovskite Solar Cells," *Advanced Materials* 34, no. 49 (2022): 2110587, https://doi.org/10.1002/adma.202110587.

18. Q. Fu, X. Tang, H. Liu, et al., "Ionic Dopant-free Polymer Alloy Hole Transport Materials for High-performance Perovskite Solar Cells," *Journal of the American Chemical Society* 144, no. 21 (2022): 9500–9509, https://doi.org/10.1021/jacs.2c04029.

19. Z. Yao, F. Zhang, Y. Guo, et al., "Conformational and Compositional Tuning of Phenanthrocarbazole-Based Dopant-Free Hole-transport Polymers Boosting the Performance of Perovskite Solar Cells," *Journal of the American Chemical Society* 142, no. 41 (2020): 17681–17692, https://doi.org/10.1021/jacs.0c08352.

20. D. Cai, N. Saengprachum, Z. Lin, and T. Qiu, "Thermophysical Properties of 4-dimethylaminopyridine Based Ionic Liquids," *Journal of Molecular Liquids* 297 (2020): 111875, https://doi.org/10.1016/j.molliq.2019.111875.

21. T. M. Schwartz, M. E. Burnett, and K. N. Green, "Electronic Influence of Substitution on the Pyridine Ring Within NNN Pincer-type Molecules," *Dalton Transactions* 49, no. 7 (2020): 2356–2363, https://doi.org/10.1039/C9DT04714J.

22. S. Winstein, E. Clippinger, A. H. Fainberg, R. Heck, and G. C. Robinson, "Salt Effects and Ion Pairs in Solvolysis and Related Reactions. III. 1 Common Ion Rate Depression and Exchange of Anions During Acetolysis2, 3," *Journal of the American Chemical Society* 78, no. 2 (1956): 328–335, https://doi.org/10.1021/ja01583a022.

23. S. E. Wheeler and J. W. G. Bloom, "Anion–π Interactions and Positive Electrostatic Potentials of N-Heterocycles Arise from the Positions of the Nuclei, not Changes in the π-Electron Distribution," *Chemical Communications* 50, no. 76 (2014): 11118–11121, https://doi.org/10.1039/C4CC05304D.

24. S. Abbate, G. Longhi, G. Mazzeo, C. Villani, S. Petkovic, and R. Ruzziconi, "CF_3: An Overlooked Chromophore in VCD Spectra. A Review of Recent Applications in Structural Determination," *RSC Advances* 9 (2019): 11781–11796, https://doi.org/10.1039/c9ra01358j.

25. M. Namatame, M. Yabusaki, T. Watanabe, Y. Ogomi, S. Hayase, and K. Marumoto, "Direct Observation of Dramatically Enhanced Hole Formation in a Perovskite-Solar-Cell Material Spiro-OMeTAD by Li-TFSI Doping," *Applied Physics Letters* 110, no. 12 (2017), https://doi.org/10.1063/1.4977789.

26. F. Lamberti, T. Gatti, E. Cescon, et al., "Evidence of Spiro-OMeTAD De-doping by Tert-Butylpyridine Additive in Hole-Transporting Layers for Perovskite Solar Cells," *Chem* 5, no. 7 (2019): 1806–1817, https://doi.org/10.1016/j.chempr.2019.04.003.

27. L. Caliò, M. Salado, S. Kazim, and S. Ahmad, "A Generic Route of Hydrophobic Doping in Hole Transporting Material to Increase Longevity of Perovskite Solar Cells," *Joule* 2, no. 9 (2018): 1800–1815, https://doi.org/10.1016/j.joule.2018.06.012.

28. H. J. Snaith and M. Grätzel, "Enhanced Charge Mobility in a Molecular Hole Transporter via Addition of Redox Inactive Ionic Dopant: Implication to Dye-sensitized Solar Cells," *Applied Physics Letters* 89, no. 26 (2006): 262114, https://doi.org/10.1063/1.2424552.

29. Y. Zhang, Z. Zhou, F. Ji, et al., "Trash into Treasure: Δ-$FAPbI_3$ Polymorph Stabilized $MAPbI_3$ Perovskite with Power Conversion Efficiency Beyond 21%," *Advanced Materials* 30, no. 22 (2018): 1707143, https://doi.org/10.1002/adma.201707143.

30. J. W. Lee, S. H. Bae, Y. T. Hsieh, et al., "A Bifunctional Lewis Base Additive for Microscopic Homogeneity in Perovskite Solar Cells," *Chem* 3, no. 2 (2017): 290, https://doi.org/10.1016/j.chempr.2017.05.020.

31. R. A. Belisle, P. Jain, R. Prasanna, T. Leijtens, and M. D. McGehee, "Minimal Effect of the Hole-Transport Material Ionization Potential on the Open-Circuit Voltage of Perovskite Solar Cells," *ACS Energy Letters* 1, no. 3 (2016): 556–560, https://doi.org/10.1021/acsenergylett.6b00270.

32. P. Caprioglio, M. Stolterfoht, C. M. Wolff, et al., "On the Relation Between the Open-Circuit Voltage and Quasi-Fermi Level Splitting in Efficient Perovskite Solar Cells," *Advanced Energy Materials* 9, no. 33 (2019): 1901631, https://doi.org/10.1002/aenm.201901631.

33. P. Caprioglio, C. M. Wolff, O. J. Sandberg, et al., "On the Origin of the Ideality Factor in Perovskite Solar Cells," *Advanced Energy Materials* 10, no. 27 (2020): 2000502, https://doi.org/10.1002/aenm.202000502.

16146840, 0, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/aenm.202506760 by Seoul National University, Wiley Online Library on [16/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

34. X. Zheng, Z. Li, Y. Zhang, et al., "Co-Deposition of Hole-Selective Contact and Absorber for Improving the Processability of Perovskite Solar Cells," *Nature Energy* 8, no. 5 (2023): 462–472, https://doi.org/10.1038/s41560-023-01227-6.

35. A. Sun, C. Tian, R. Zhuang, et al., "High Open-Circuit Voltage (1.197 V) in Large-Area (1 cm^2) Inverted Perovskite Solar Cell via Interface Planarization and Highly Polar Self-Assembled Monolayer," *Advanced Energy Materials* 14 (2024): 2303941, https://doi.org/10.1002/aenm.202303941.

36. H.-S. Yun, Y.-H. Seo, C.-E. Seo, et al., "Surface Engineering of Tin Oxide Nanoparticles by pH Modulation Facilitates Homogeneous Film Formation for Efficient Perovskite Solar Modules," *Advanced Energy Materials* 14, no. 25 (2024): 2400791, https://doi.org/10.1002/aenm.202400791.

37. H. Kim, D. Y. Lee, J. Lim, et al., "Enhancing Stability and Efficiency of Perovskite Solar Cells with a Bilayer Hole Transporting Layer of Nickel Phthalocyanine and Poly(3-Hexylthiophene)," *Advanced Energy Materials* 13, no. 35 (2023): 2301046, https://doi.org/10.1002/aenm.202301046.

38. M.-H. Li, X. Ma, J. Fu, et al., "Molecularly Tailored Perovskite/Poly (3-hexylthiophene) Interfaces for High-Performance Solar Cells," *Energy & Environmental Science* 17, no. 15 (2024): 5513–5520, https://doi.org/10.1039/D4EE02251C.

39. Q. Lou, X. Xu, Z. Xu, et al., "A Two-in-One Additive Strategy for Poly (3-hexylthiophene)-Based Perovskite Solar Cells with Improved Efficiency and Stability," *Cell Reports Physical Science* 6, no. 4 (2025): 102515, https://doi.org/10.1016/j.xcrp.2025.102515.

40. Q. Cheng, W. Chen, Y. Li, and Y. Li, "Recent Progress in Dopant-Free and Green Solvent-Processable Organic Hole Transport Materials for Efficient and Stable Perovskite Solar Cells," *Advanced Science* 11, no. 17 (2024): 2307152, https://doi.org/10.1002/advs.202307152.

41. Y. Kim, G. Kim, E. Y. Park, et al., "Alkylammonium Bis (trifluoromethylsulfonyl) Imide as a Dopant in the Hole-transporting Layer for Efficient and Stable Perovskite Solar Cells," *Energy & Environmental Science* 16, no. 5 (2023): 2226–2238, https://doi.org/10.1039/D2EE04045J.

42. E. H. Jung, N. J. Jeon, E. Y. Park, et al., "Efficient, Stable and Scalable Perovskite Solar Cells Using Poly (3-hexylthiophene)," *Nature* 567, no. 7749 (2019): 511–515, https://doi.org/10.1038/s41586-019-1036-3.

43. C. A. R. Perini, E. Rojas-Gatjens, M. Ravello, et al., "Interface Reconstruction From Ruddlesden–Popper Structures Impacts Stability in Lead Halide Perovskite Solar Cells," *Advanced Materials* 34, no. 51 (2022): 2204726, https://doi.org/10.1002/adma.202204726.

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